Exercise Sheet

Task 1 – Flipping Bits in Assembly

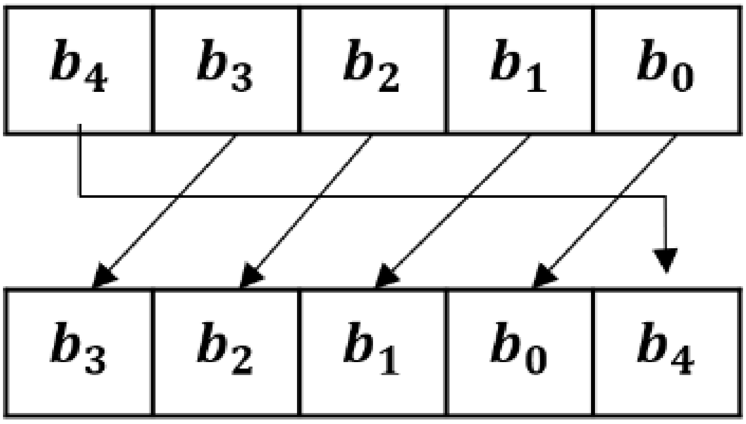
Write an assembly program that flips the LSB of the value saved in register t0. Use the documentation provided in the Moodle course.
What do you have to change to flip the second last bit?

Task 2 – Two's Complement in Assembly

Write a program to compute the two's complement representation of a number. You can assume, that the amount of said number is already in register t0. You do not have to implement an overflow check at this time. Use the documentation provided in the Moodle course.

Task 3 – Rotations

You already learned what shifts are in the lecture. Here we want to perform a rotation, a shift where the bit is pushed out at one end and then inserted back into the register at the other. There are no RISC V assembly instructions to perform rotations, therefore it will be your task to write a short program that implements the functionality of a logical rotation to the left, as can be seen here:



Task 4 – Instruction Formats

How big is the address space for jump and branch instructions? Which address space is bigger?